



The Hypersensitive Fluorescence Analyzer

User Manual

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Thank you for choosing Wellray's hypersensitive fluorescence analyzer. Please read this manual in detail to understand the operation method of the equipment.

1 Product Introduce

1.1 Name and model

◆ Name: Hypersensitive Fluorescence Analyzer

◆ Model: WR-1808/1608D

1.2 Technical principle

This device is to detect the fluorescence intensity of the lanthanide fluorescent microsphere tracer marker in the immune specific reaction product. Based on the detected fluorescence value, i.e. standard curve of marker's biological concentration, the device uses cloud computing to immediately convert and quantitatively/semiquantitative determine the specific reactant's concentration, composition and other information in the tested object.

1.3 The range of application

This equipment is suitable for the detection of fluorescent immunodiagnostic reagents. It is widely used in the field of medicine, biology, animal and plant diseases, food safety, feed safety, drugs, environmental protection and other fields. It has realized rapid, quantitative (semiquantitative) detection in the field and laboratory.

1.4 Product parameters

- | | |
|---|--|
| ◆ Electric source: DC 5V-1A (Adapter or USB power supply) | ◆ N.W. : ≤ 0.5 kg |
| ◆ Size: 15cm×9.5cm×5.5cm | ◆ Operation mode: Continuous operation |

1.5 Work environment

- | | |
|---|---------------------------------------|
| ◆ Environment temperature: 5°C~40°C | ◆ Environmental humidity: $\leq 80\%$ |
| ◆ Far away from high-strength electromagnetic field | ◆ Avoid strong light |

1.6 The main technical indicators

- | | |
|--|--------------------------------|
| ◆ Linear range: related coefficient(R) $\geq 99\%$ | ◆ Repeatability: CV $\leq 5\%$ |
| ◆ Stability: Relative offset $\leq \pm 5\%$ | ◆ Detection time $\leq 8s$ |

1.7 Equipment configuration

Number	Product Name	Quantity
1	Hypersensitive Fluorescence Analyzer	One
2	Data line	One
3	Instructions	One

2 Instructions

2.1 Application requirement

- Requires grouped or customized testing reagents.
- Needs to operate in conjunction with "cloud detection" APP or PC software.

2.2 Power source

2.2.1 Power Supply

Connecting one end of the USB cable to the device and the other end to the power supply.

Note: you can use regular mobile phone power adapter, power bank or any other USB interface that provides stable power supply satisfying 5V 1A.



2.2.2 Turn on /off the device

Press the button on the front of the device for two seconds and release it. The indicator light will be on and the device will start up.

Press and hold this button for two seconds, then release it, and the indicator light will be off, and the device will power off.



2.3 Software installation and device connection

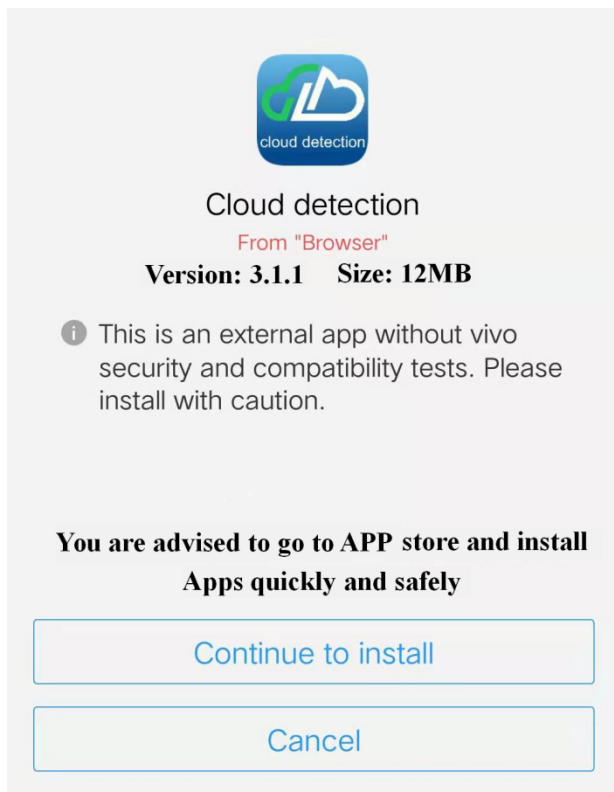
2.3.1 Install "cloud detection" App

1) Using your own smart terminal, please follow the steps below to self-install the APP software: use WeChat or other software to scan the QR code on the right to download. Or, go to Wellray's official website (<https://www.wellray.net>) to download.

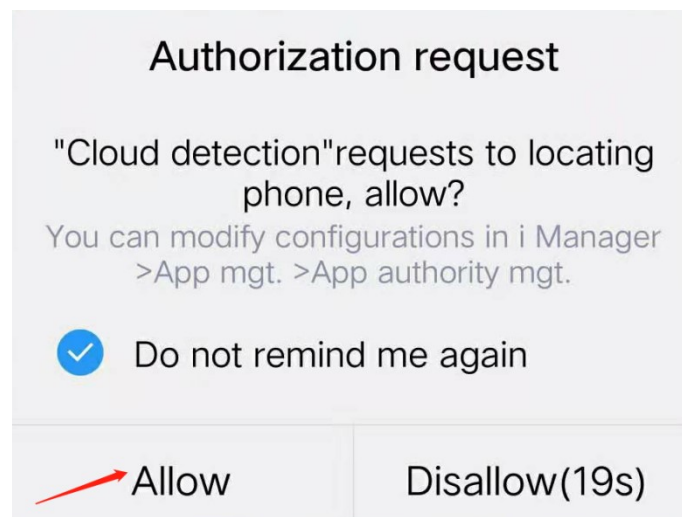
2) After downloading, the smart terminal will automatically



enter the software installation interface and click "Install". In case of this security caution, please click "continue to install".



3) When the software runs for the first time, it will request for location and Bluetooth permission, please be sure to allow.



Note: the software is connected to the device via Bluetooth, please keep the Bluetooth function of the mobile phone on when using the device.

2.3.2 Registered account

1) Open "Cloud Detection" App software on the phone, click "Sign up", fill in the device SN code, your email, verification code, password and other information to register.

You can log in your personal account at multiple terminals and synchronize data at multiple terminals.

The image shows two side-by-side screenshots of the app's login and registration screens.

Log in screen: Features a blue header with a back arrow and the text "Log in". Below the header is a grey box with a user icon and the text "We focus on your data security". The main form has two input fields: "Username" with the placeholder "Please enter your email" and "Password" with the placeholder "Please enter your password". There is a checkbox labeled "Remember password" which is checked. A blue "LOG IN" button is at the bottom. At the very bottom, there are links for "Forget password" and "Sign up" (with a red arrow pointing to it).

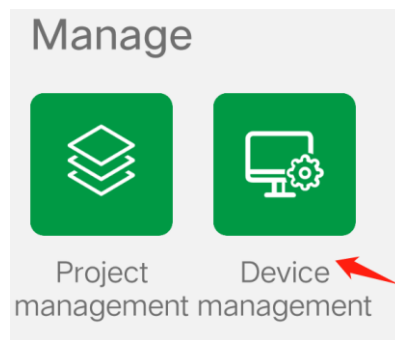
Sign up screen: Features a blue header with a back arrow and the text "Sign up". The form includes several fields with red asterisks indicating required information: "Enter or scan the device SN" (with a QR code icon), "Please input your email", "Verification code" (with a "GET VERIFICATION CODE" button), "Password", "Confirm your password", and "Nickname". A blue "SIGN UP" button is at the bottom.

2) After registration, the user can go to the section "Me" to modify the account's personal information in the "Edit" section, to change data synchronization setting in the "Privacy" option, to report issues about the software in the "Feedback" option, to share test data in the "Share to" option, to update software version in the "About" option, and/or to sign out the account.

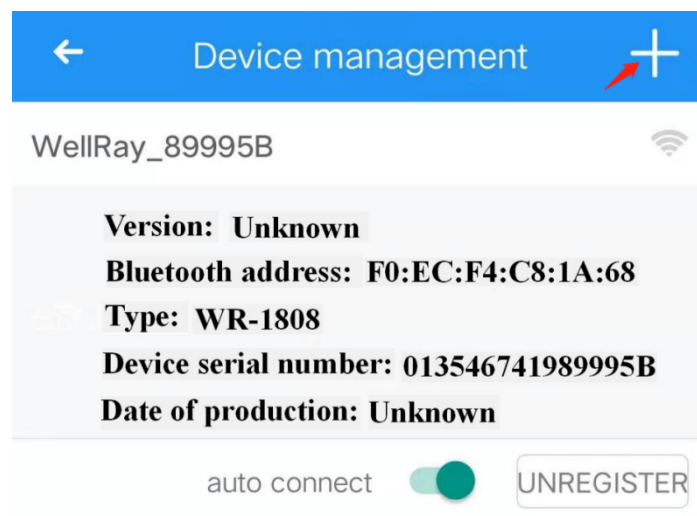
The image shows the "Me" (Profile) screen of the app. It has a blue header with the text "Me" and a plus icon. Below the header is a grey box with a user icon and the name "Mei". The main content area lists several options, each with an icon and a right-pointing arrow: "Edit" (pencil icon), "Privacy" (lock icon) with the text "Data sync is turned on", "Feedback" (speech bubble icon), "Share to" (share icon), and "About" (info icon) with the text "Check for updates". At the bottom of this list is a grey box with a "Sign out" button. The bottom navigation bar has two items: "Work platform" with a folder icon and "Me" with a person icon (highlighted in blue).

2.3.3 Connect device

- 1) Keep the device on, log in to mobile App "Cloud Detection", and click "Device management".



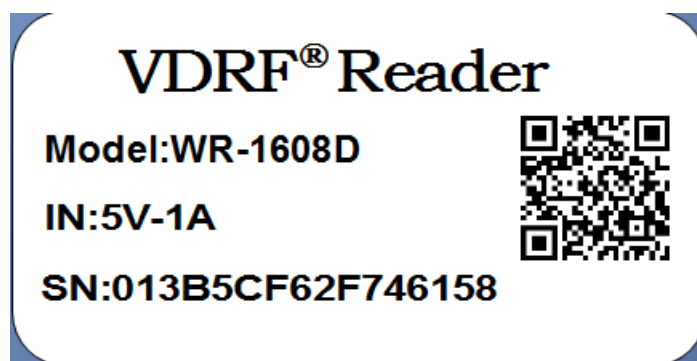
- 2) For the first time the device is used to establish a connection or to repair linked devices, please click the icon in the upper right corner to enter the device's search page. (Device management functions records the last device you used and automatically connects to it.)



- 3) Select the device you are going to use, click the device number to make a pairing connection (If the list is blank, please check whether the device is powered on).



Note: the device number corresponds to the last six digits of SN code at the rear end (or bottom) of the device.



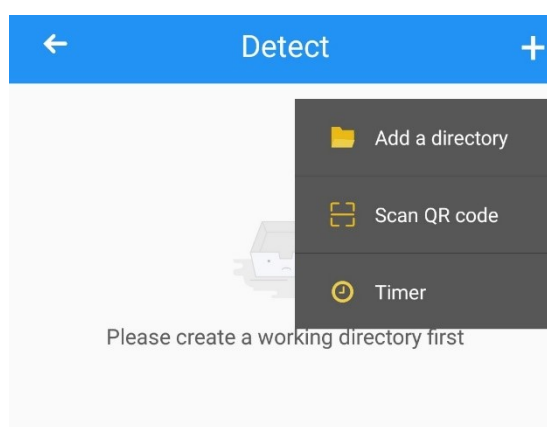
2.4 Testing

2.4.1 Create working directory

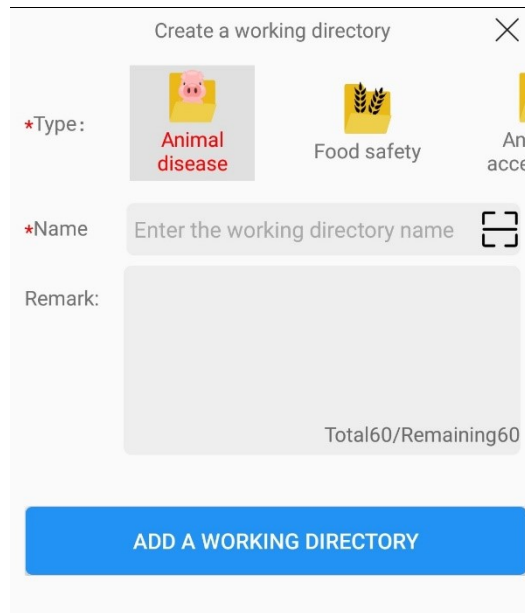
- 1) In the main menu interface, find and click "Detect" to enter the detection main interface.



- 2) Click to expand the icon in the upper right corner to create a working directory (take "universal detection" as an example).



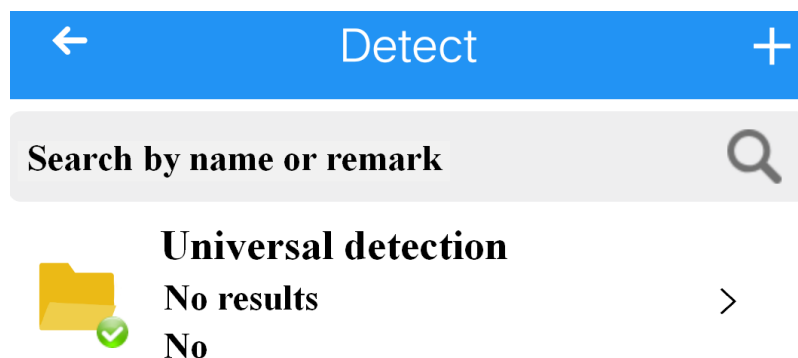
3) Enter the working directory creation interface, fill in the corresponding information, and click the button of "ADD A WORKING DIRECTORY". After adding successfully, the software will return to the working directory list interface.



The dialog box titled "Create a working directory" contains the following fields and options:

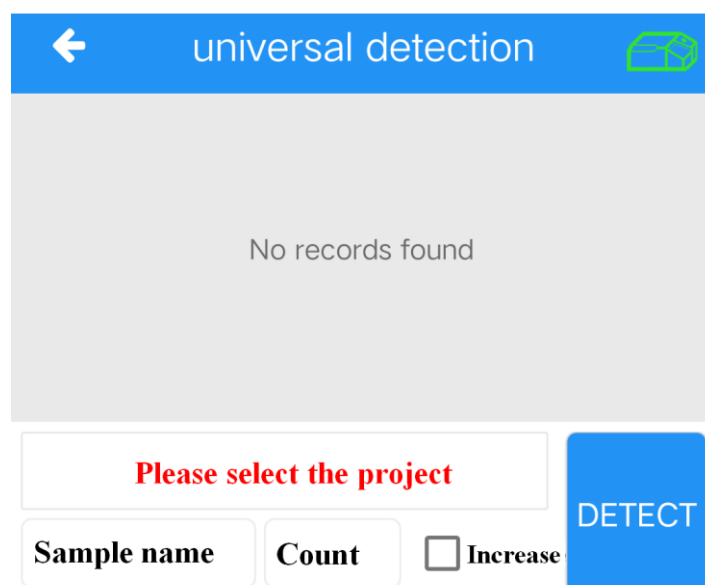
- *Type:** Three icons are shown: "Animal disease" (a pig icon), "Food safety" (a wheat icon), and "Antimicrobial resistance" (a pill icon).
- *Name:** A text input field with the placeholder "Enter the working directory name".
- Remark:** A large text area for additional notes.
- Count:** A label "Total60/Remaining60" is located at the bottom right of the remark area.
- Action:** A blue button at the bottom labeled "ADD A WORKING DIRECTORY".

The new working directory:



2.4.2 Select test projects

1) Click the new working directory to enter the detection interface.



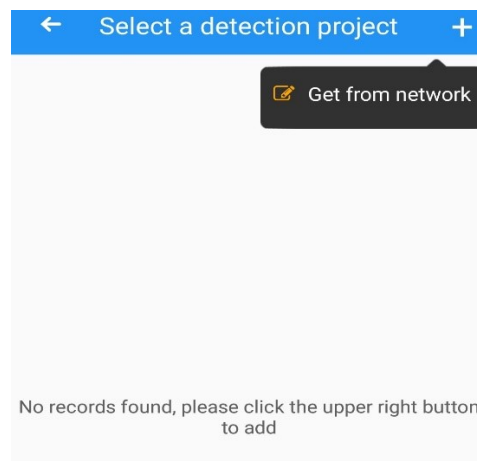
The "universal detection" screen has a blue header with a back arrow, the title "universal detection", and a folder icon. The main area is a large grey rectangle with the text "No records found". At the bottom, there is a red prompt "Please select the project", a table with headers "Sample name" and "Count", a checkbox labeled "Increase", and a blue "DETECT" button.

2) Click "Please select the project" to enter the project list interface, where you can directly select existing projects or add new test projects.

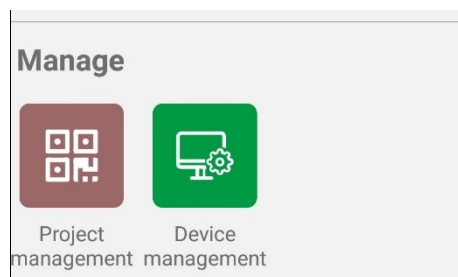
Note: the selected projects must be consistent with the test kit.



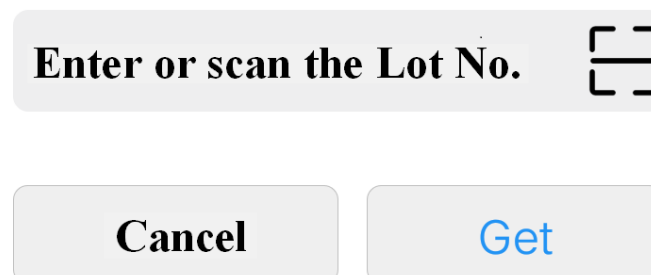
3) Add test projects: click the "+" button on the upper right of the "Select a detection project", and click "Get from network".



Note: Test project can also be added by clicking the "project management" icon on the main menu page.



4) Click the scan button in the pop-up interface to enter the scan interface, scan the QR code on the side of the test kit, or manually enter the complete serial number.



Serial number and QR code are on the side of the kit, as shown in the figure:

P.D.:		POX-Ab
G.P.: 12 months		POX19073016576
Storage: Light-shielding, hermetic,		P.D.: 2019.07.30
refrigeration (2-8) preserves.		
Mfrs.: Chengdu Wellray Bio-technology Co.,Ltd.		
Add.: No.26, Xixin Avenue, Gaoxin West District,		
Chengdu.		
Tel.: 028-81496879		
P.C.: 611731		
Web.: www.wellray.net		
WeChat Official Account: WeiRuiBio		

5) Click on "Get" to preview the test project's information.

POX19073016576	
Cancel	Get

6) After confirming that the test project is correct, click the "ADD TO" button at the bottom of the project information interface to complete the add. Return to the test project list interface and select the test project just obtained.

		Project management	
Search by project name or Lot No.			
	POX-Ab	<i>just</i>	
	Object sheep		
	Lot No. POX19073016576		

7) Click the name of the test project to return to the test interface. The test project has been selected. Then enter the sample number. If you select "Increase", the serial number will automatically increase by 1 after each test.

POX-Ab POX19073016576		DETECT
AG1635	1 ☒ Increase	

2.4.3 Test sample

1) After adding samples and conducting chromatography according to the instructions, insert the reagent card to be tested into the device, close the light-shielding cover, and click the "detect" button in the lower right corner of the screen (at this time, you will hear the sound of "tick" from the device, indicating that the device has received the test command from the software and started to test the reagent card).



2) After the test, the test results will be displayed in the working directory, as shown in the following figure:

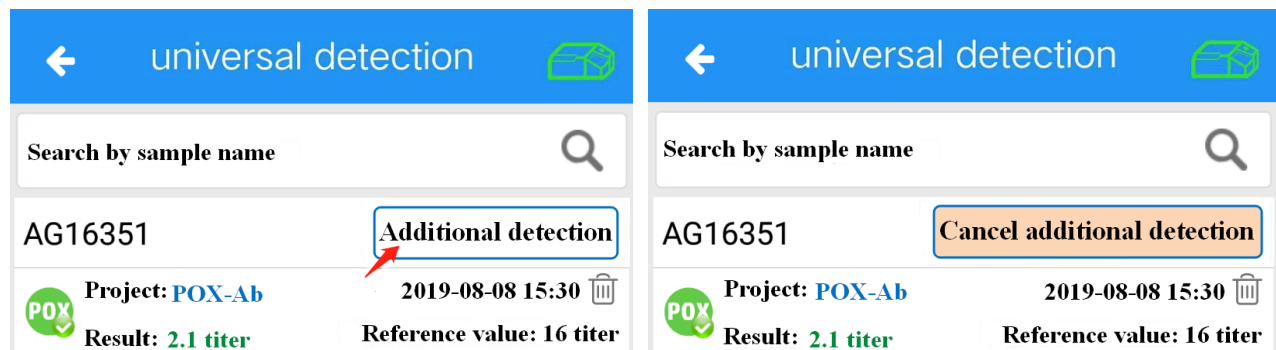
← universal detection		
Search by sample name 		
AG16351	Cancel additional detection	
	Project: POX-Ab Result: 2.1titer	2019-08-08 15:30  Reference value:16 titer
PD3427	Additional detection	
	Project: POX-Ab Result: 3.7titer	2019-08-03 15:34  Reference value:16 titer

Note: Please strictly follow the instructions to use reagent card. The paper instructions are in the kit.

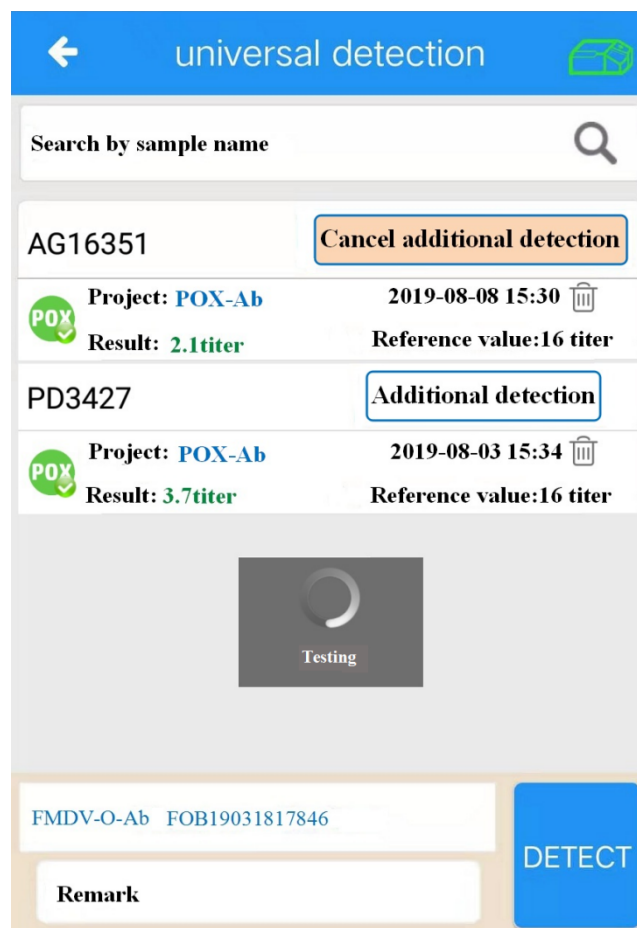
2.4.4 Additional detection

The same sample can be tested multiple times at different time for different test projects. The software automatically classifies and summarizes the test results, which is convenient for data comparison and analysis. The operation steps are as follows:

- 1) Click the "Additional detection" button at the right of the sample name to add another test project on the same sample. The option button will switch to the "Cancel additional detection" status, and the page will enter the additional test mode.



- 2) Select a test project, and input remark information, click "DETECT" button to start the test.



3) After the test, the test results will be automatically displayed under the sample name. At this time, you can repeat step 2 to continue the detection.

← universal detection

Search by sample name

AG16351

Cancel additional detection

Project: PPRV-Ab

2019-08-15 09:30

Result: 2.6 titer

Reference value: 10 titer

Project: FMDV-O-Ab

2019-08-13 11:34

Result: 4.0 titer

Reference value: Pig 64 titer, cattle and sheep 128 titer

Project: POX-Ab

2019-08-08 15:30

Result: 2.1 titer

Reference value: 16 titer

PD3427

Additional detection

Project: POX-Ab

2019-08-03 15:34

Result: 3.7 titer

Reference value: 16 titer

Please select the project

Remark

DETECT

4) When the additional detection is being conducted, you cannot create a new sample number and test. After the additional detection is completed, click "Cancel additional detection" to exit the additional detection mode.

← universal detection 

Search by sample name 

AG16351

Additional detection

PRB

Project: PPRV-Ab

2019-08-15 09:30 

Result: 2.6 titer

Reference value:10 titer

FOB

Project: FMDV-O-Ab

2019-08-13 11:34 

Result: 4.0 titer

Reference value:
Pig 64 titer, cattle and sheep128 titer

POX

Project: POX-Ab

2019-08-08 15:30 

Result: 2.1 titer

Reference value:16 titer

PD3427

Additional detection

POX

Project: POX-Ab

2019-08-03 15:34 

Result: 3.7 titer

Reference value:16 titer

Please select the project

Sample name

Count

☐ Increase

DETECT

1

2.4.5 View test results

Click the test result to view the test result details. When additional detection is being conducted, the user needs to "cancel additional detection" to view the details.

 Detail



PD3427

Test result 3.7 titer

Reference value 16 titer

Project POX-Ab

Operation time 2019-08-08 15:30:32

Lot No. POX19073016576

Device No. 0133C4B88E9964BC

Detection object Sheep

Chromatography time 15 minutes

Dilution multiple 50

Range 0-2048 titer

Description Negative: the antibody titer $\leq$ 16
Positive: the antibody titer $>$ 16

LAST

NEXT

3 Precautions and maintenance and service

3.1 Precautions

- 1) Please read the operation manual carefully before using the device and operate according to the manual.
- 2) This device is a portable optical precision instrument. Please handle it with care and be careful not to fall.
- 3) In order to ensure the accuracy of the test data, each device has been strictly adjusted and calibrated before delivery. If any problem is found, please contact hotline service and return the equipment for maintenance. Please don't dismantle it by yourself.
- 4) The biohazard sign “” at the insertion of the test card indicates that the operator should wear protective gloves when detecting serum, plasma and residual toxic substances.
- 5) After testing, reagent cards and other wastes should be handled in accordance with the relevant regulations of medical waste management in medical and health institutions.

3.2 Maintenance and service

- 1) Please try to keep the environment clean when operating the device.
- 2) Take out the reagent card in time after the test to avoid corrosion of the device.
- 3) If the device is not used for a long time, please place it in a dry and clean environment.

4 Troubleshooting and maintenance

- 1) In case of device failure, please refer to the following "common troubleshooting methods".
- 2) If the issue cannot be resolved, please contact the company's hotline service department and describe the issue to get timely support.

If your device needs to be returned to the factory for maintenance, in order to reduce the impact on your work, the company will provide standby device for your normal use as soon as possible. Common troubleshooting methods:

Trouble	Solutions
Device can't connect	Check and turn on Bluetooth
Connection lost	Check whether the power supply of the equipment is stable
Unable to start the device, or detection is interrupted without reason.	Under voltage of power supply. Please use a power supply no less than 5V 1A
Device making abnormal sound	Close the light-shielding cover before testing
	Please insert the reagent card correctly (Pay attention to both sides)
	Insert the reagent card in place
	Check whether the reagent card shell has foreign matter.

The software prompt abnormal	You can operate according to the prompts
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5 Transportation and storage

5.1 Transportation

The equipment shall be handled with care during transportation. Avoid violent vibration. Prevent excessive moisture.

5.2 Storage conditions

- ◆ Temperature: -10~40°C
- ◆ Humidity: 20%~90%
- ◆ Atmospheric pressure: 86~106kPa
- ◆ Non-corrosiveness gas
- ◆ Environmental drying

【Ordering Information】



Hangzhou DaRay Technology Co., Ltd

Room 201, Building 3, No. 9 Jiankang Road, Tonglu County, Hangzhou, Zhejiang, China 311599

Tel:+86-028-81496879

Fax:+86-028-81496879

<https://www.wellray.net>